

Coding Exercise: Analysis of Steady-State Visually Evoked Potentials (SSVEP)

Goals

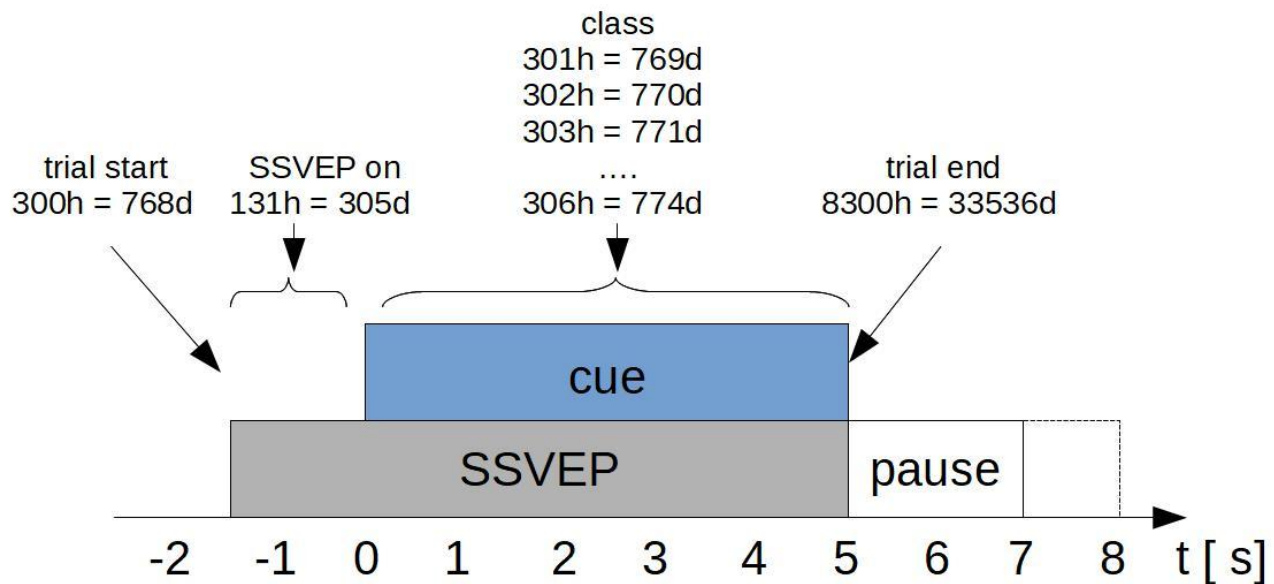
This exercise is designed to:

- Provide an example of analyzing SSVEP, one of the most popular BCI paradigms
- Although SSVEP still represents evoked responses (i.e., like the error-potential ERPs examined in BCI Lab 4), in this case analysis in the frequency domain is required and exemplified.
- Provide additional insights on signal processing, feature extraction and machine learning methods used in BCI.

Description

You are given a dataset of a single subject consisting of 4 EEG files stored in BDF format. Each file corresponds to a single run/block of a protocol designed to generate SSVEP EEG responses. Each file corresponds to a single execution of the protocol containing 36 trials. Signals were collected at a sampling rate of 2048 Hz. The EEG data were recorded at 64 scalp locations according to the 10-20 system. The channel labels and order can be retrieved from the header of the loaded BDF files and data can be loaded with the `bdfmatlab` toolbox as already explained in the context of Lab 2. All events are stored in the trigger channel of the loaded file as pulses. The trigger channel is in the last channel index, specifically, at position 89 of the overall channel set. This set includes more than the 64 EEG channels (total 89, other channels regard unused EOG, EMG and other auxiliary channels). The EEG channels can be retrieved in the first 64 positions as denoted by the channel labels in the header. The position (timing) of each event can be found with command: `gettrigger(triggerchannel, 305)`. In this protocol, each trial started with two visual flickering stimuli on either side of a fixation cross in the middle of the screen. One second into the trial, the user would see an arrow cue instructing him/her to stare at the left or right flickering stimulus. The instructed stimulus is the target stimulus of each trial (the information of which stimulus was used in each trial as non-target/distractor is not saved). There are 6 types of stimuli, flickering at frequencies 7.5 Hz, 9 Hz, 10 Hz, 12 Hz, 15 Hz and 20 Hz. In each BDF file, each of the 6 stimuli is the target stimulus for 6 trials (i.e., there are 36 total trials in each file). The corresponding cue events take on values 769, 770, 771, 772, 773, 774, respectively. The start of each trial, when the two stimuli start flickering on either side (equivalently: 1 second before the cue),

are marked with the event 768. The trial timeline can be seen in the figure below. The neurophysiological data can be downloaded from this link: [Zenodo link](#)



Tasks

1. Perform the trial extraction procedure to isolate as “trials” the segments of EEG where the subject stares at a specific stimulus. Make sure you maintain also the information of which stimulus was the target stimulus in each trial (ground truth “label”).
2. Inspect the data and decide whether you need any kind of temporal/spectral filtering and apply this/these accordingly. In particular, are there drifts in the data? If so, how could you eliminate those? Which techniques can do this, and which one would you prefer and why? If you choose to use some low-/high-/band-pass filter, what consideration should you take into account given the nature of this protocol?
3. Apply CAR and/or Laplacian spatial filtering. Inspect the grand average plots and feature fitness maps produced in the next steps and decide whether the choice of spatial filter (or the absence of any such filter) is critical for this protocol. After you have accomplished the decoding/classification task, compare the performance without spatial filtering, with CAR and with Laplacian.
4. Reflect on whether the information in the EEG of this protocol (evoked responses of the brain to flickering stimuli) will be more easily located in the time or frequency domain. Simply said, do we expect particular waveforms elicited by the flickering visual stimuli? Or do we expect modulation of power in particular bands of the signal spectrum? Check the book regarding different approaches to acquire spectral information of the signal and implement at least two of these.

5. Produce grand averages of the above SSVEP correlates for all channels, separately for each stimulus. Plot these grand averages for different channels (make educated guesses given your knowledge on the SSVEP paradigm. Is the theory verified regarding which cortical region undertakes visual processing and resonates to flickering stimuli?
6. Given these grand averages, do you think the choice of all flickering frequency has been wise? If not, why?
7. Do the different types of spectral features you have attempted agree on the most responsive channels? Are there any differences? Is the grand average enough for concluding which feature extraction method is the best? Enrich the grand average plots with the missing info using the function `shadedErrorBar` to extract safer conclusions.
8. Is the frequency resolution provided by the spectrum extraction methods adequate? Is it maybe unnecessarily high in terms of spectral resolution? How can you deal with the latter?
9. Try to isolate the most responsive channel/band pairs. What kind of methods do you know that can be used for this task? Perform this process with one or more different algorithms. Which method should be more suitable for this and why? Produce heatmaps (feature fitness maps) to visualize these results.
10. Using the feature fitness maps you have acquired from the previous step, produce topoplots that show clearly which EEG channels hold most of the information regarding SSVEP responses.
11. Perform classification with one or more classification models of your choice using supervised learning. Use training/testing split or cross-validation to avoid overfitting. How can you avoid selecting features associated to only some, and not all, of the 6 stimuli present in the protocol?
12. Is the accuracy obtained satisfactory? Is performance balanced across different classes (i.e., stimuli)? If certain stimuli are confused with each other, can you tell why that could be? Connect this with the question above on whether the choice of flickering frequencies was appropriate in this protocol.
13. Using unsupervised learning in this problem will necessitate a calibration session before any closed-loop BCI can be realized. Reflect whether in this case going through this cumbersome procedure is absolutely necessary. Can you propose a principle or a method for avoiding BCI calibration in SSVEP?